Interpretable by Construction, Identifiable by Construction: Privacy Leakage of Shapelet-Based Anti-Money-Laundering Models

Anonymous Author(s)

ABSTRACT

Anti-money-laundering (AML) models have to explain their alerts, and shapelet classifiers do this by pointing to a short pattern in an account's activity. Most shapelet methods take these patterns straight from training accounts, and the strongest one also stores the class of the account each pattern came from. We ask what a party that receives such a model learns about the bank's customers. On two AML benchmarks and on real bank data, a party that already holds transaction histories for the bank's customers can link released patterns back to individual customers even when every amount it holds is off by up to 5%, with 96–98% precision, and so learn which of them the bank had flagged. Revealing that is what tipping-off rules forbid. The link breaks when the party sees only half of each account's transactions. Aggregate membership audits, LiRA included, barely register the problem, while checking each released pattern on its own does. Learned shapelets, which are not copied from any account, XX, and a differentially private version keeps most of their accuracy on two of three datasets. Prototype networks leak in the same way. Example-based explanations in AML need their own privacy review.

CCS CONCEPTS

• **Security and privacy** → **Privacy protections**; • **Computing methodologies** → *Machine learning*; • **Applied computing** → Economics.

KEYWORDS

anti-money laundering, shapelets, explainability, membership inference, re-identification, tipping off

1 INTRODUCTION

Banks increasingly use machine learning to decide which accounts to send to AML investigators, and supervisors expect each alert to come with a reason. Deployed systems usually pair boosted trees with post-hoc explanations [7]. Example-based models offer another route, because the explanation is a piece of training data. Prototype networks [14] work this way, and so do shapelets. Ye and Keogh [1] introduced shapelets as short subsequences whose distance to a series decides its class, so the model can show which pattern triggered a decision. The shapelet transform [2] is part of HIVE-COTE 2.0 [11], RDST [4] is among the strongest single time-series classifiers, and shapelets have been used for early classification [12] and financial forecasting [13]. We know of no published AML deployment. Laundering typologies such as structuring and fan-in/fan-out are shapes in time, though, so shapelets are a natural candidate wherever regulators ask for explanations.

Most shapelet methods *select* their patterns by cutting them out of training series. RDST also stores, for every pattern, the normalisation statistics and the class of the account it came from. *Learned* shapelets [3] are instead optimised by gradient descent from random starting values. It is already known that example-based explanations reveal training records [18], and that card-transaction traces are so distinctive that a handful of points single out most people [19]. A copied window of someone's account activity will therefore usually point to one person. We confirm this (82–97% of non-empty RDST shapelets match exactly one account), and treat it as the starting point. The questions that matter in practice are who can actually make the link, how much error in their data they can tolerate, what they learn, and whether the usual privacy audits notice.

Tipping off. AML law forbids telling anyone that a customer has been reported or investigated for suspicious activity (FATF Recommendation 21 [20]; 31 U.S.C. §5318(g)(2) [21]; EU Directive 2015/849, Art. 39 [22]; UK POCA 2002 s.333A [23]). Training labels in AML come from exactly such findings. A model that stores a recognisable piece of a customer's history together with that customer's label can therefore disclose the finding to whoever receives the model. Boosted trees and other conventional models do not create this risk in the same way. A membership signal says that an account was used for training, and the training set contains both classes.

Contributions. (i) A threat model, tied to tipping-off rules, for three parties: one holding customer data, one holding only the model, and one seeing only alert explanations. (ii) Measurements on two AML benchmarks and real bank data showing where linkage works and where it stops, including noisy and incomplete adversary data and a quarter-million candidate accounts. (iii) A comparison of audits: aggregate membership inference and LiRA miss the leak, and a per-pattern check catches it. (iv) Mitigations tested on the strongest leaker, with LiRA on learned and differentially private shapelets. (v) Evidence that prototype networks behave the same way.

2 RELATED WORK

Production AML monitoring relies on boosted trees over account aggregates with post-hoc explanations [7], and public benchmarks include SAML-D [6] and AMLworld [9]. Membership inference [16] is the standard audit of training-data leakage, and strong attacks are judged by their true-positive rate at low false-positive rates [17]. Shokri et al. [18] show that model explanations leak membership and that example-based explanations reveal training points directly. De Montjoye et al. [19] show that four transactions, known only by shop and day, re-identify 90% of card holders. Federated shapelet learning [8] and privacy-aware matrix profiles [10] protect shared patterns, but neither measures what a released model reveals about individuals. We add the AML setting, where the stored label is itself regulated information, and an audit of who can exploit the leak.

3 THREAT MODEL

A bank trains a shapelet detector on labelled accounts, where the labels come from past suspicious-activity findings. We consider three parties. *Data holder.* The model leaves the bank, for example to a vendor, an auditor, a regulator's sandbox or a consortium partner, or it is stolen. This party holds transaction histories for many of the bank's customers but not the labels or the training set. By matching shapelets to its own records it learns which customers were in the training data and, through RDST's stored class, which were flagged. We test it with an exact copy of the data and with a degraded copy: amounts perturbed, transactions missing, and only part of the customer list. *Model holder.* The party has the model and no data. From RDST it can rebuild raw windows (values × std + mean), i.e. exact weekly amounts and counts for specific but unnamed customers, which it could link later. *Explanation recipient.* The model stays in the bank, but alert explanations reach people outside the AML team, such as outsourced investigators or a vendor's case-management tool. Whether this leaks depends on the interface. Showing the stored shapelet itself shows another customer's window. Showing the alerted account's own matching segment does not. Naming the customer behind a shown window still needs account data.

4 EXPERIMENTAL SETUP

Data. *SAML-D* [6] has 9.5M synthetic transactions over 46 weeks with 28 typologies. Each account becomes a six-channel weekly series (log outgoing and incoming amount, outgoing and incoming count, cross-border amount, cash amount). Accounts involved in any laundering transaction are positive (7,902). We sample 22,044 normal accounts matched on the number of active weeks, about 2.8 per positive, because unmatched normal accounts are mostly single-week accounts that make detection trivial. *AMLworld* HI-Small [9] covers about ten days, so we use 6-hour bins (40 steps, 23,409 accounts, 5,862 positive). *Berka* [15] holds real, anonymised accounts of a Czech bank from 1993–1998 (4,500 accounts, 72 monthly steps). It has no laundering labels, so we predict which customers hold a loan (682 positive) after removing all loan repayments. The privacy measurements do not depend on the task. Each dataset is split 50/50 into members (training) and non-members, stratified by label, and results average 3 seeds.

Detectors. CatBoost on 40 aggregates plus the flattened series stands in for industry practice. We also train MiniRocket [5] with logistic regression; a classic shapelet transform (200 shapelets chosen from 5,000 random z-normalised candidates by class separation, i.e. AUC instead of information gain, with self-similar candidates removed) followed by ExtraTrees; RDST (aeon, 1,000 shapelets) followed by ExtraTrees; and learned shapelets [3] (144, soft-min distance) followed by ExtraTrees.

Attacks. A shapelet *singles out* a customer if exactly one candidate account contains a window within 10^{-3} of it. The *degraded adversary* rebuilds its series from raw transactions with log-normal amount noise (σ=0.05, about ±5%), keeps each transaction with probability 0.8 or 0.5, and may hold only part of the customer list. It names a customer only if the nearest account is at least twice as close as the second nearest (300 sampled RDST shapelets per seed). For *bank scale* we add 220k further SAML-D accounts with at least five transactions and re-test 150 shapelets that were unique among the study accounts. For *membership* we use the aggregate score (−minimum distance to any shapelet, or confidence for the baselines), LiRA [17] with 32 shadow models on Berka, and a per-pattern audit that flags an account when it contains a released shapelet that matches no other account.

Table 1: A data holder with imperfect data (SAML-D, 300 sampled RDST shapelets per seed, mean over seeds). Its copy may have perturbed amounts and missing transactions, and may cover all or half of the customers. It names a customer only when the nearest account is at least twice as close as the second nearest. Cust.: customers named correctly. Prec.: share of names that are correct.

	All accounts		Half of accounts	
Adversary's view	Cust.	Prec.	Cust.	Prec.
Exact copy	251	94%	125	86%
Amounts ±5%	227	96%	115	86%
80% of txns, ±5%	71	78%	36	62%
50% of txns, ±5%	10	27%	5	17%

Table 2: SAML-D, exact copy of the data (mean ± std over 3 seeds). PR-AUC and recall at 1% FPR on held-out accounts; MIA AUC is the aggregate membership audit; Re-ID is the share of released shapelets that match exactly one of 29,946 accounts.

Model	PR-AUC	R@1%FPR	MIA AUC	Re-ID (%)
CatBoost (aggregates)	0.983 ± 0.001	0.842 ± 0.031	0.504 ± 0.002	n/a
MiniRocket	0.955 ± 0.002	0.637 ± 0.044	0.508 ± 0.003	n/a
Shapelet transform	0.878 ± 0.007	0.493 ± 0.033	0.500 ± 0.002	15.0 ± 2.2
RDST	0.948 ± 0.006	0.631 ± 0.021	0.520 ± 0.001	85.6 ± 1.6
RDST, norm.-only, no stats	0.925 ± 0.005	0.515 ± 0.015	0.512 ± 0.000	74.9 ± 1.9
Learned shapelets	0.957 ± 0.004	0.667 ± 0.017	0.500 ± 0.000	0.0

5 RESULTS

Who can link a pattern to a customer (Table 1). Amount noise of ±5% barely slows the data holder. With 300 RDST shapelets it makes 227 correct identifications on SAML-D at 96% precision, 62 of them of flagged customers. On Berka's real accounts it makes 207 at 98% precision. With half the customer list it finds about half as many people, still mostly correctly. Missing activity is what stops it. When it sees 80% of each account's transactions it still makes 71 correct identifications at 78% precision on SAML-D, and with 50% its precision falls to 27%. The threat is real for parties that see an account's full activity, such as vendors and auditors given bank data, insiders, or anyone holding breached data, and weak for outsiders with a partial view.

What exact matching adds (Tables 2 and 3). With an exact copy of the data, 86% of RDST shapelets single out one account on SAML-D, about 827 customers per model, of whom 217 were flagged. This mostly restates that transaction windows are unique,

Table 3: AMLworld HI-Small (23,409 accounts, 6-hour bins) and Berka (4,500 real accounts, monthly bins, loan task), mean over seeds. MIA: aggregate membership AUC. Re-ID: share of shapelets that match exactly one account. Cust.: customers singled out per model.

Model	PR-AUC	MIA AUC	Re-ID (%)	Cust.
AMLworld				
CatBoost	0.619 ± 0.004	0.625 ± 0.004	n/a	n/a
MiniRocket	0.475 ± 0.003	0.586 ± 0.003	n/a	n/a
Shapelet transform	0.408 ± 0.016	0.499 ± 0.002	6.0 ± 1.1	–
RDST	0.571 ± 0.001	0.507 ± 0.002	64.5 ± 1.2	621 ± 10
RDST, norm.-only	0.546 ± 0.005	0.498 ± 0.002	40.4 ± 1.7	391 ± 14
Learned shapelets	0.389	0.500	0.0	0
Berka (real)				
CatBoost	0.583 ± 0.013	0.632 ± 0.009	n/a	n/a
MiniRocket	0.625 ± 0.012	0.561 ± 0.003	n/a	n/a
Shapelet transform	0.470 ± 0.014	0.502 ± 0.002	43.5 ± 2.1	–
RDST	0.421 ± 0.021	0.530 ± 0.003	75.4 ± 0.8	637 ± 6
RDST, norm.-only	0.391 ± 0.016	0.501 ± 0.004	50.2 ± 1.0	445 ± 17
Learned shapelets	0.357 ± 0.017	0.499 ± 0.001	0.0	0

Table 4: LiRA placeholder XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX.

a	b	c	d	e
x	1	2	3	4
x	1	2	3	4
x	1	2	3	4
x	1	2	3	4
x	1	2	3	4

Table 5: Mitigations applied to RDST (mean over seeds; AMLworld seed 0). Each defence rewrites the released windows, after which the detector is refit. Link: among shapelets that singled out their source before the change, the share whose nearest account is still that source (chance is below 0.1%).

	SAML-D		AMLworld		Berka	
Release	PR-AUC	Link	PR-AUC	Link	PR-AUC	Link
As trained	0.946	100	0.571	100	0.420	100
Noise σ=0.1	0.937	71	0.557	84	0.420	70
Noise σ=0.25	0.935	64	0.549	73	0.409	35
Noise σ=0.5	0.931	53	0.540	65	0.396	14
Noise σ=1	0.925	24	0.530	43	0.389	3
Mean of k=5	0.945	44	0.568	43	0.422	33
Mean of k=20	0.945	35	0.568	28	0.419	17
Mean of k=50	0.944	34	0.567	29	0.424	11

and the same holds on AMLworld and on Berka's real accounts (75%, about 637 of 4,500 customers). Among sampled unique shapelets, 100% stay unique when the candidate list grows to 249,946 accounts. The more telling numbers concern shape alone. The classic transform keeps only z-normalised shapes and still singles out its source in 15% of cases on SAML-D and 44% on Berka. If RDST is forced to normalise every shapelet and its statistics are withheld, 75% of shapelets still single out one customer at a small accuracy cost (PR-AUC 0.925 vs. 0.948), because a six-channel dilated shape is itself a fingerprint. A model holder who keeps the statistics recovers a median of 13 non-zero weekly amounts and counts per identifying shapelet.

Audits (Table 4). The few hundred source accounts are a small share of all members, so the aggregate membership AUC on SAML-D is 0.52. On Berka, LiRA on the RDST shapelets reaches only X.X% true positives at 1% false positives. Checking each released pattern on its own flags XXX members and X non-members on Berka (827 and 0 on SAML-D). Because Berka has only 2,250 non-members, we report rates at 1% false positives, which rest on 22 accounts. Auditors of example-based models should test patterns one by one. Conventional models leak in a different way. CatBoost shows no membership signal on SAML-D but overfits on AMLworld and Berka (aggregate AUC 0.62 on AMLworld), and LiRA on Berka reaches X.X% at 1% false positives. That tells an attacker an account was used for training. It does not reveal the account's label or its transactions.

Explanations (Appendix Table 9). If an interface explains each alert by showing the three most important RDST shapelets that occur in the account, every alert on all three datasets shows another customer's exact window. This follows from how RDST works, so the more useful finding is its shape. The same 16 SAML-D customers appear across all 3,951 alerts, each in about 242 unrelated cases, together with their recorded class. Showing the alerted account's own matching segment instead avoids the problem at no cost.

Prototype networks (Appendix Table 8). A ProtoPNet-style network [14] projects each prototype onto its nearest training patch so that it can be shown. 92–98% of projected prototypes match only their source customer on all three datasets, and projection costs no accuracy.

Mitigations (Table 5). We apply each defence to RDST, the strongest leaker. Neither cheap defence removes the link. Gaussian noise costs accuracy before it helps. At σ=1, one channel standard deviation, PR-AUC falls from 0.946 to 0.925 on SAML-D and from 0.571 to 0.530 on AMLworld, and 24% and 43% of the shapelets that singled out their source still have it as their nearest account. Replacing each shapelet by the mean of its k nearest training windows keeps accuracy, but even at k=50 about a third still lead back to the source on the synthetic data (11% on Berka). The source's own window is one of the k nearest, and the average stays closest to it. Learned shapelets do not copy any account: none lies within 10^{-2} of an account on SAML-D, and pushing them to memorise a small training set (Appendix Table 7) produces no confident identification. LiRA confirms this on Berka. XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX. Their accuracy depends on the data. On SAML-D they are the best shapelet model (PR-AUC 0.957 vs. 0.948 for RDST), on AMLworld they trail RDST (0.39 vs. 0.57), and on Berka they reach 0.36 vs. 0.42. Training them with DP-SGD bounds what the shapelets and their linear head reveal. At ε=1 PR-AUC is 0.947 on SAML-D, unchanged on AMLworld, and 0.29 on Berka. XX XX XX XX XX XX XX XX XX XX XX XX

XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX.

6 DISCUSSION AND LIMITATIONS

Scope. Banks may not use shapelets for AML today. We study them as the clearest case of an explanation that is a stored training record plus its label, and prototype and case-based models share that property. **Data.** The AML benchmarks are synthetic and their positives are true laundering cases rather than the investigator reports a bank would train on, and Berka uses a proxy task. Sampling about three normal accounts per positive makes detection easier than at real prevalence, but adding 220k accounts did not dilute re-identification. **Methods.** Our classic transform scores candidates by AUC because a standard implementation broke down on our sparse series, and RDST is the off-the-shelf aeon implementation. LiRA was run on Berka only. The per-pattern audit is built around this leak on purpose. Our point is that general-purpose audits miss it. **Law.** Whether a given release counts as tipping off depends on jurisdiction and circumstances; UK s.333A, for instance, requires that the disclosure be likely to prejudice an investigation. We point to the risk and make no claim about liability. **Defences.** Differentially private shapelet *selection* remains open.

Ethics. SAML-D and AMLworld are synthetic. Berka is a public, anonymised research release, and we report only aggregate statistics.

Recommendations. Treat released shapelets and the labels stored with them as confidential AML information. Do not rely on normalisation or on withholding statistics. In explanation interfaces, show the alerted account's own segment rather than the stored pattern. Prefer learned shapelets, with DP-SGD where a guarantee is needed, and check the accuracy cost on your own data. Audit example-based models pattern by pattern, at low false-positive rates.

REFERENCES

[1] L. Ye and E. Keogh. 2009. Time series shapelets: a new primitive for data mining. In *KDD*.

[2] J. Hills, J. Lines, E. Baranauskas, J. Mapp, and A. Bagnall. 2014. Classification of time series by shapelet transformation. *Data Mining and Knowledge Discovery* 28.

[3] J. Grabocka, N. Schilling, M. Wistuba, and L. Schmidt-Thieme. 2014. Learning time-series shapelets. In *KDD*.

[4] A. Guillaume, C. Vrain, and W. Elloumi. 2022. Random dilated shapelet transform: a new approach for time series shapelets. In *ICPRAI*.

[5] A. Dempster, D. F. Schmidt, and G. I. Webb. 2021. MiniRocket: a very fast (almost) deterministic transform for time series classification. In *KDD*.

[6] B. Oztas, D. Cetinkaya, F. Adedoyin, M. Budka, H. Dogan, and G. Aksu. 2023. Enhancing anti-money laundering: development of a synthetic transaction monitoring dataset. In *IEEE ICEBE*.

[7] P. Tertychnyi, M. Godgildieva, M. Dumas, and M. Ollikainen. 2022. Time-aware and interpretable predictive monitoring system for anti-money laundering. *Machine Learning with Applications* 8.

[8] Z. Liang and H. Wang. 2024. FedST: secure federated shapelet transformation for time series classification. *The VLDB Journal* 33.

[9] E. Altman, J. Blanuša, L. von Niederhäusern, B. Egressy, A. Anghel, and K. Atasu. 2023. Realistic synthetic financial transactions for anti-money laundering models. In *NeurIPS Datasets and Benchmarks*.

[10] L. Zhang, J. Ding, Y. Gao, and J. Lin. 2023. PMP: privacy-aware matrix profile against sensitive pattern inference for time series. In *SIAM SDM*.

[11] M. Middlehurst, J. Large, M. Flynn, J. Lines, A. Bostrom, and A. Bagnall. 2021. HIVE-COTE 2.0: a new meta ensemble for time series classification. *Machine Learning* 110.

[12] Z. Xing, J. Pei, P. S. Yu, and K. Wang. 2011. Extracting interpretable features for early classification on time series. In *SIAM SDM*.

[13] J. Kim, H. Lee, H. Jeon, S. Jin, and S. Ko. 2025. From patterns to predictions: a shapelet-based framework for directional forecasting in noisy financial markets. arXiv:2509.15040.

[14] C. Chen, O. Li, D. Tao, A. Barnett, C. Rudin, and J. K. Su. 2019. This looks like that: deep learning for interpretable image recognition. In *NeurIPS*.

[15] P. Berka. 1999. Guide to the financial data set. In *PKDD'99 Discovery Challenge*.

[16] R. Shokri, M. Stronati, C. Song, and V. Shmatikov. 2017. Membership inference attacks against machine learning models. In *IEEE S&P*.

[17] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr. 2022. Membership inference attacks from first principles. In *IEEE S&P*.

[18] R. Shokri, M. Strobel, and Y. Zick. 2021. On the privacy risks of model explanations. In *AIES*.

[19] Y.-A. de Montjoye, L. Radaelli, V. K. Singh, and A. Pentland. 2015. Unique in the shopping mall: on the reidentifiability of credit card metadata. *Science* 347, 6221.

[20] FATF. 2012 (updated 2023). *International Standards on Combating Money Laundering and the Financing of Terrorism & Proliferation*, Recommendation 21: Tipping-off and confidentiality.

[21] 31 U.S.C. §5318(g)(2). Reporting of suspicious transactions: prohibition on notification of disclosures; see also 31 CFR 1020.320(e).

[22] Directive (EU) 2015/849 of the European Parliament and of the Council, Article 39 (prohibition of disclosure).

[23] Proceeds of Crime Act 2002 (UK), section 333A: tipping off in the regulated sector.

Table 9: Explanations that show the stored shapelet (RDST, mean over seeds). The top N test accounts are alerted, where N is the number of test positives. Each alert is explained by three shapelets that occur in it, either the most important ones (*global*) or the best-matching among the 50 most important (*local*). We count alerts whose explanation contains another customer's exact window, and how many different customers are shown.

	SAML-D	AMLworld	Berka
Alerts explained	3,951	2,931	341
Alerts showing another customer, global (%)	100	100	100
Customers exposed, global	16	18	13
of which flagged, global	4	5	2
Alerts showing another customer, local (%)	60	38	100
Customers exposed, local	33	29	18
of which flagged, local	5	9	2

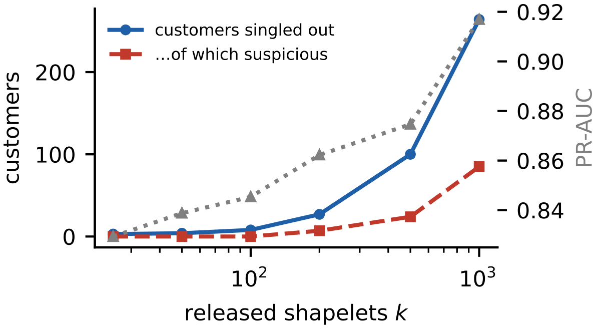

Figure 1: Classic shapelet transform on SAML-D, seed 0: customers singled out (and how many were flagged) grow with the number of released shapelets k, while accuracy levels off.

A ADDITIONAL RESULTS

Table 6: Mitigations applied to the classic shapelet transform (SAML-D, mean over seeds). Unique: the shapelet matches only its source. Top-1: the source is the nearest account.

Release	PR-AUC	Unique (%)	Top-1 (%)
As trained	0.878 ± 0.007	15.0 ± 2.2	19.3 ± 1.6
Noise σ=0.1	0.879 ± 0.007	0.0 ± 0.0	10.8 ± 2.5
Noise σ=0.25	0.877 ± 0.007	0.0 ± 0.0	3.2 ± 0.5
Noise σ=0.5	0.876 ± 0.005	0.0 ± 0.0	0.2 ± 0.2
Noise σ=1.0	0.870 ± 0.004	0.0 ± 0.0	0.0 ± 0.0
k-anon. mean, k=5	0.877 ± 0.007	0.3 ± 0.2	13.5 ± 0.4
k-anon. mean, k=20	0.876 ± 0.005	0.2 ± 0.2	12.5 ± 2.1
k-anon. mean, k=50	0.876 ± 0.006	0.0 ± 0.0	10.7 ± 1.3

Table 7: Learned shapelets trained hard on small sets to encourage memorising (SAML-D, seed 0). Nearest is member: share of shapelets whose nearest account is a training member, with the members' share of all accounts in brackets. Claims: confident identifications (second-nearest account at least twice as far).

Training	PR-AUC	Nearest-is-member (chance)	Claims
30 epochs, full train	0.957	50% (50%)	0
200 epochs, full train	0.952	50% (50%)	0
200 epochs, 1k train	0.928	4% (3%)	0
400 epochs, 300 train	0.895	0% (1%)	0

Table 8: Prototype networks and DP-SGD learned shapelets (mean over 3 seeds). ProtoNet: a ProtoPNet-style network whose prototypes are projected onto training patches; Re-ID is the share of projected prototypes that match only their source customer. DP: learned shapelets trained with DP-SGD ($\delta = 1/n$) at $\varepsilon = \infty$, 8 and 1.

	SAML-D	AMLworld	Berka
ProtoNet PR-AUC	0.959	0.479	0.338
ProtoNet Re-ID (%)	92	98	93
Customers exposed / 20	16.7	18.3	17.7
DP $\varepsilon = \infty$ PR-AUC	0.961	0.398	0.357
DP $\varepsilon = 8$ PR-AUC	0.954	0.414	0.288
DP $\varepsilon = 1$ PR-AUC	0.947	0.412	0.288

Table 6 gives the mitigations on the classic transform. Table 7 trains learned shapelets on small subsets for many epochs to push them towards memorising, and Table 8 reports prototype networks and DP-SGD learned shapelets. The DP model at ε=∞ uses the training loop of the DP model and its own head, so its PR-AUC differs slightly from the learned shapelets in Tables 2 and 3. On AMLworld all learned variants sit near 0.40 and the noise of DP-SGD acts as a mild regulariser. Table 9 details the explanation experiment, where the *local* variant shows the best-matching shapelets among the 50 most important ones.